Sahaj Bhadja

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Summary

Client-focused Software Engineer and USC MSCS graduate with 2+ years building **real-time financial systems**, distributed services, data platforms, and reliable full-stack applications across ambiguous business requirements.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Translated stakeholder requirements into a multi-tenant platform with normalized schemas, row-level isolation, and indexed **real-time position queries**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users.
- Architected asynchronous Python services with explicit job states, retries, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **mission-critical system reliability** across 5K+ monthly workflows.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Owned React and Flask workflows from requirements through release, supporting 150+ stakeholders and 10K+ daily requests through **client-focused delivery**, REST APIs, production debugging, and iterative feedback cycles.
- Designed concurrent Python services for secure file ingestion and multimodal processing, handling 1K+ files per batch with validation, provider fallbacks, and **automated integration testing** across production workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed cross-platform integration modules using object-oriented patterns and REST APIs, improving response times 30%, reducing technical debt 20%, and strengthening **maintainable system architecture** across distributed application services.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **technical leadership**, code reviews, defect resolution, and dependable releases across Windows, Linux, and macOS.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight microservices for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **real-time WebSocket workflows**, REST APIs, validation controls, and responsive React interfaces across distributed locations.
- Developed TypeScript integrations and transaction-safe record workflows, while Jenkins CI/CD, Docker staging, and **automated regression testing** reduced deployment time 50% and improved delivery confidence across controlled production releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built React and TypeScript transaction workflows for scheduling, reservation tracking, pricing, and payment-status updates, delivering **sub-200-millisecond financial APIs** across 50K+ service records and multiple client-facing demonstration reviews.
- Designed data models and integration contracts for 12+ endpoints, translating ambiguous stakeholder requirements into technical architecture while reducing **upstream design errors** 25% before implementation, planning, and delivery.

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C++, C#, Bash, HTML5, CSS3
Capital Markets: Position Keeping, Trade Lifecycle, Settlements, Reconciliation, Securities Lending, Collateral, Liquidity, Market Risk, P&L, Multi-Currency Processing, Financial Data Modeling
Systems & Data: Spring Boot, FastAPI, Flask, React, Node.js, REST APIs, Microservices, Kafka, WebSockets, PostgreSQL, SQL Server, MongoDB, Redis, ETL, Event-Driven Architecture, Distributed Systems
Delivery: AWS, Azure, Docker, Kubernetes, Git, GitHub, CI/CD, Jenkins, Unit Testing, Integration Testing, Code Review, Debugging, Agile, SDLC, System Design, Requirements Analysis, Project Planning, Technical Documentation, Stakeholder Management

Projects

Global Position Keeper | *Java, Spring Boot, Kafka, PostgreSQL, Redis, WebSockets*

- Designed an event-driven platform that maintains real-time securities and cash positions across accounts, currencies, and markets, using **idempotent trade processing**, reconciliation controls, streaming updates, and configurable business rules.

Settlement Control Tower | *Python, FastAPI, Kafka, SQL, React, Docker*

- Built a settlement workflow application that models trade obligations, flags delivery exceptions, and coordinates resolution through **auditable state transitions**, operational dashboards, client-defined rules, resilient integrations, and automated tests.

FinNexus | *Next.js, TypeScript, PostgreSQL, Kubernetes, Plaid, WebSockets*

- Built a financial platform with Plaid account connectivity, real-time WebSocket updates, responsive TypeScript workflows, and **transactional PostgreSQL data models**, deploying containerized services through Kubernetes for reliable financial operations.